

ChickpeaOmicsR

This package provides gene expression data for chickpea under different stress conditions. Several Data MayBe Loaded to the Environment. Gene Expression counts are normalized. Alsamman M. Alsamman, aalsamman100[at and not .]gmail[now .]com.

Installation

This packages will installed automatically when you install ChickpeaOmicsR. But it can be installed separately as well to make things easier.

```
library(devtools)

## Loading required package: usethis
#install.packages("devtools")
library(igraph)

##
## Attaching package: 'igraph'

## The following objects are masked from 'package:stats':
##
##      decompose, spectrum

## The following object is masked from 'package:base':
##
##      union

#install.packages("igraph")
library(pheatmap)
#install.packages("pheatmap")
library(ggplot2)
#install.packages("ggplot2")
library(dplyr)

##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:igraph':
##
##      as_data_frame, groups, union

## The following objects are masked from 'package:stats':
##
##      filter, lag

## The following objects are masked from 'package:base':
##
##      intersect, setdiff, setequal, union

#install.packages("dplyr")
```

You can install the released version of ChickpeaOmicsR from CRAN with:

```
library(ChickpeaOmicsR)

## Welcome to ChickpeaOmicsR!

## [1] "This package provides gene expression data for chickpea under different stress conditions."
## [1] "Several Data MayBe Loaded to the Environment."
```

```
## [1] "Gene Expression counts are normalized"
## [1] "Alsamman M. Alsamman, aalsamman100[at and not .]gmail[now .]com"
#devtools::install_github("AlsammanAlsamman/ChickpeaOmicsR")
```

Datasets

ChickpeaOmicsR package provides several datasets that can be used to perform various analyses and visualizations. The datasets are as follows:

1. **CaExpressionCounts**: Gene expression counts for chickpea. Generated from RNA-seq data of Gene expression counts for 28891 gene and 197 samples.
2. **CaExpressionMetadata**: Metadata associated with gene expression studies.
3. **CaGenomeAnnotation**: Genome annotations of chickpea genes.
4. **CaProteinEnrichment**: Protein enrichment analysis of all Chickpea genes.
5. **CaProteinSequenceInfo**: Protein sequence information.
6. **CaProteinInteractionNetwork**: Protein-protein interaction data.
7. **CaExpressionSignificance**: Data on significant gene expression changes.
8. **CaGWAS**: Genome-wide association study data for chickpea.

These datasets can be used to perform various analyses and visualizations within the package.